MAT367
Intro to Differential Geometry

Class Lecture Notes

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Class Lectures

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Preface

These notes were created during class lectures. As such, they may be incomplete or lacking in some detail at parts, and may contain confusing typos due to time-sensitivity. Additionally, these notes may not be comprehensive. Most statements in this document which are not Theorems, Problems, Lemmas, Corollaries, or similar, are likely paraphrased to a certain degree. Please do not treat any material in this document as the exact words of the original lecturer.

If you are viewing this document in Obsidian, you may notice that the links in the pdf document do not work. This is intentional behaviour, as I currently do not have or know of a decent solution which allows them to behave well with the setup in Obsidian. However, below certain pages, there may be links to other documents - these are usually context-relevant links between notes of different areas of study. I created these links to point out potential similarities, or in case one area of study is borrowing a concept, definition, or theorem from another area of study, and you wish to see the full, original definition/derivation/proof or whatever it may be.

IV Tangent Spaces and Vector Fields

7 Tangent Spaces

Lec 11 - Feb 10 (Week 6)

recall: if $M \subseteq \mathbb{R}^n$ an embedded mfld, the **tangent space** at $p \in M$ is T_pM given as:

$$T_pM \;=\; \{\text{velocity vectors of curves } \gamma : (a,b) \longrightarrow M \text{ at } 0 \text{ where } \gamma(0) = p\} \subseteq \mathbb{R}^n$$

this is in fact an m-dim'l vspace. indeed, M^m is locally the img of some $\varphi : U \subseteq \mathbb{R}^m \longrightarrow \mathbb{R}^n$, so $T_pM = D\varphi\big|_{\varphi^{-1}(p)}$ where $p \in M$. note φ is not a chart map here. by chain rule, given $v \in \mathbb{R}^n$, take $\gamma(t) = \varphi^{-1}(p) + tv$. then:

$$\frac{\mathrm{d}}{\mathrm{d}t}(\varphi \circ \gamma)\big|_0 \;=\; D\varphi_p\left(\frac{\mathrm{d}\gamma}{\mathrm{d}t}\bigg|_0\right) \;\implies\; D\varphi\big|_{\varphi^{-1}(p)}(\mathbb{R}^m) \subseteq T_pM$$

to get the reverse inclusion, start with $\gamma : (a,b) \longrightarrow M$ and use $\varphi^{-1} \circ \gamma$. ok idt that was all that important tbh

using directional derivatives, we motivate the defn for an abstract mfld.

Definition 7.1

a **tangent vector** at $p \in M$ is a linear map $v : C^\infty(M) \longrightarrow \mathbb{R}$ of the form:

$$v(f) \;=\; \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} f(\gamma(t))$$

where $\gamma : (a,b) \longrightarrow M$ is some smooth curve with $\gamma(0) = p$.

the idea is to use curves γ to take dir derivatives of f at a pt p. in this sense, each fixed v represents a direction along which we can take the dir derivative of a fn. this identifies it (implicitly, later explicitly) with a particular vector in M which represents the dir more manifestly. this distinct identification separates the direction and its behaviour on fns with the vector in M itself; in this sense, they are really two ways of writing the same thing. tangent spaces drawn in $\mathbb{R}^n$ are thus "geometric representations" of them.

remarks:

1. this is a subset of the inf-dim $\mathrm{Hom}(C^\infty(M), \mathbb{R})$
2. can see it as an equiv rel on curves; that is, v is an equiv class of γ inducing the same linear map. more precisely, curves γ_1, γ_2 are equiv if $\gamma_1(0) = \gamma_2(0) = p$, and $\gamma_1'(0) = \gamma_2'(0)$. v is then the equiv class, and v intuitively represents the vec $\gamma_i'(0)$.

in other words, want to think of them not as vectors but as dir derivatives. we provide an alt coord-wise defn.

Definition 7.2

let $p \in M, (U, \varphi)$ chart arnd p. $v : C^\infty(M) \longrightarrow \mathbb{R}$ a **tangent vector** at p if it is a linear combo of functionals in $\mathrm{Hom}(C^\infty(M), \mathbb{R})$:

$$f \mapsto \sum_{i=1}^{m} a_i \partial_i \left(f \circ \varphi^{-1}\big|_{\varphi(p)} \right)$$

for some $a \in \mathbb{R}^m$, where ∂_i is the i-th partial deriv.

here, $a \in \mathbb{R}^m$ represents the dir vector mapped to coords in $\mathbb{R}^m$. remarks:

1. by defn, is a m-dim'l linear subspace of $\mathrm{Hom}(C^\infty(M), \mathbb{R})$
2. not obv that it doesnt depend on (U, φ)

Theorem 7.3

the two defn's are equiv.

Proof. Source: Primary Source Material

given v, let $\widetilde{\gamma}(t) = \varphi(p) + ta$ a curve $I \to \varphi(U)$ and $\gamma = \varphi^{-1} \circ \gamma$. then:

$$v(f) = \sum_{i=1}^{m} a_i \partial_i \left(f \circ \varphi^{-1}\big|_{\varphi(p)} \right) = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} (f \circ \varphi^{-1})(\widetilde{\gamma}(t)) = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} f(\gamma(t))$$

by chain rule, so v of the [first] form. now, if v [of the first form] for some γ w $\gamma(0) = p$, then:

$$v(f) \;=\; \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} (f \circ \varphi^{-1})(\varphi \circ \gamma(t)) \;=\; \nabla(f \circ \varphi^{-1})\bigg|_{\varphi(p)} \cdot \frac{\mathrm{d}}{\mathrm{d}t}(\varphi \circ \gamma)\bigg|_{t=0}$$

letting $a := \frac{\mathrm{d}}{\mathrm{d}t}(\varphi \circ \gamma)\Big|_{t=0}$, this is then of the second form. ■

ah. so, here we're identifying $v = \sum a_i \partial_i$, or in other words, vectors a with "operators" v. thus, v represents a directional deriv in the direction of a. in the first defn, a is given as the derivative of γ at $t = 0$. <- previous ramblings

ok now onto the txtbk. this is gonna suck.

using the 2nd defn, any chart U arnd p clearly gives an iso $T_pM \simeq \mathbb{R}^m$ by identifying v with a. specifically, if $U \subseteq M$ open, then $T_pU \subseteq C^\infty(U)$ is the subspace spanned by the partials at p:

$$\frac{\partial}{\partial x_1}\bigg|_p, \;\ldots\;, \frac{\partial}{\partial x_m}\bigg|_p$$

this basis identifies $T_pU \simeq \mathbb{R}^m$. coords a_i can then be obtained by taking $v(\pi_i)$.

there is yet another characterization which is coord-free and doesn't rely on curves. first, note that clearly every v satisfies the product rule, also known as the Leibniz rule:

$$v(fg) \;=\; v(f)g(p) + f(p)v(g)$$

evidently, there are many $v \in \mathrm{Hom}(C^\infty(M), \mathbb{R})$ which do not satisfy this property. however, we claim the following:

Theorem 7.4

a linear $v : C^\infty(M) \longrightarrow \mathbb{R}$ defines an elem of T_pM iff it satisfies the product rule.

the pf is rly long and uses "Hadamard's Lemma", i highly doubt its important. im skipping it. anyway, it gives us this third defn:

Definition 7.5

given $v : C^\infty(M) \longrightarrow \mathbb{R}$, $v \in T_pM$ iff:

$$v(fg) \;=\; v(f)g(p) + f(p)v(g)$$

for all f, g.

there is *yet another*, which seemingly involves dual spaces. the book warns to skip it tho, esp if you dont like abstractions. ill read it, but im not taking notes on it.

notice that "velocity vectors" are naturally elements of the tangent space. given a curve γ, then $\gamma'(t_0) \in T_{\gamma(t_0)}M$ is given by:

$$(\gamma'(t_0))(f) \;=\; \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t_0} f(\gamma(t))$$

Lec 12 - Feb 12 (Week 6)

today: derivatives of smooth maps $f : M \longrightarrow N$. idea: $D_pf : T_pM \longrightarrow T_{f(p)}N$ should generalize the jacobian. recall that for $v \in T_pM$, $v : C^\infty(M) \longrightarrow \mathbb{R}$ is a tangent vec iff it satisfies the product rule, iff:

$$v(g) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} g(\gamma(t))$$

for a curve γ.

Definition 7.6

we define the tangent map $D_pf : T_pM \longrightarrow T_{f(p)}N$ of f at p as:

$$D_pf(v)(g) \;=\; v(g \circ f)$$

where $g \in C^\infty(N)$, and $g \circ f \in C^\infty(M)$.

this essentially takes a tangent vector $v \in T_pM$ and "pushes" it thru f to get a tangent vector $D_pf(v) \in T_{f(p)}N$. p straightforward.

Proposition 7.7

an equiv defn is that $D_p f([\gamma]) = [f \circ \gamma]$. acc this js shows the above is well-defn'd.

proof:

$$\begin{aligned} D_p f(v)(h) \;=\; v(h \circ f) \;&=\; \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} (h \circ f)(\gamma(t)) \\ &=\; \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} h(f \circ \gamma(t)) \\ &=\; [f \circ \gamma] \end{aligned}$$

notationally, we can write $Df_p(v)$ as $f_* v$, called the **pushforward** of v. we can denote by f^* a **pullback** by f as in post-composition. if we write $f^* g = g \circ f$ as the pullback of a fn g, $f_* \gamma = f \circ \gamma$ as the pushforward of the curve γ, and $f_* v = (T_p f)(v)$ the pushforward of the tangent vector v, then indeed the defining eq'n becomes:

$$(f_* v)(g) \;=\; v(f^* g)$$

chain rule is written as $D_p(g \circ f)(v) = D_{f(p)} g(D_p f(v))$. proof: symbol push. also, for $f : M \longrightarrow N$, do it on coords. now, we redefine stuff from last wk using tangent spaces.

- the **rank** of f at p is the rank of Df_p
- f is a **submersion** if Df_p is surjective, an **immersion** if it is injective
- $p \in M$ is a **critical pt** of f if $\operatorname{rank} Df_p < \min(m, n)$, a **regular pt** otw
- $q \in N$ is a **regular value** of f if all pts in $f^{-1}(q)$ are regular

given $S \subseteq M$ submfld, the inclusion map ι induces injections $D\iota_p : T_p S \hookrightarrow T_p M$.

example: given $f : M \longrightarrow N$ and reg val q, $S := f^{-1}(q)$ a codim n submfld. we claim that $T_p S = \ker Df_p$. proof: note they have same dim, so enough to prove one inclusion. since $f \circ \iota \equiv q$, take derivatives to get $T_p f(T_p \iota) = 0$.

example: G is a matrix lie grp. lmao okay. this is a subgrp + submfld of $M_n(\mathbb{R})$. then $T_I G \subseteq M_n(\mathbb{R})$ is called the **lie algebra** of G, denoted $\mathfrak{g}$. torture that g on the rack

fact: $\mathfrak{g}$ closed under $[X, Y] = XY - YX$, i.e. the commutator of all X, Y in $\mathfrak{g}$. this is the lie bracket of X, Y.